

Computer Networks

A Complete Reference Guide

From fundamentals to advanced concepts

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1. What is a Computer Network?

A **computer network** is a collection of interconnected devices (computers, servers, phones, printers) that can communicate and share resources. The internet is simply the world's largest computer network.

- **Resource sharing** — printers, files, databases
- **Communication** — email, messaging, video calls
- **Centralised data management** — servers, cloud storage
- **Distributed computing** — workload across many machines

2. The OSI Model

The OSI (Open Systems Interconnection) model is a 7-layer conceptual framework for how network communication works. Data travels **down** the stack on the sender (each layer adds a header — *encapsulation*) and **up** the stack on the receiver (each layer strips its header — *decapsulation*).

All People Seem To Need Data Processing

(Application → Physical)

Layer 7 — Application: User-facing services. **Protocols:** HTTP, HTTPS, FTP, SMTP, DNS, DHCP, SSH. **PDU:** Data

Layer 6 — Presentation: Data translation, encryption, compression. **Protocols:** SSL/TLS, JPEG, ASCII. **PDU:** Data

Layer 5 — Session: Session management between applications. **Protocols:** NetBIOS, RPC. **PDU:** Data

Layer 4 — Transport: End-to-end delivery, flow control. **Protocols:** TCP, UDP. **PDU:** Segment

Layer 3 — Network: Logical addressing and routing. **Protocols:** IP, ICMP, OSPF, BGP, ARP. **PDU:** Packet

Layer 2 — Data Link: Node-to-node frame delivery using MAC addresses. **Protocols:** Ethernet, Wi-Fi. **PDU:** Frame

Layer 1 — Physical: Raw bits over physical medium (copper, fibre, radio). **PDU:** Bit

3. TCP/IP Model

While the OSI model is theoretical, the **TCP/IP model** (4 layers) is what the internet actually uses.

TCP/IP Layer	↔	OSI Layers
Application	↔	7 Application, 6 Presentation, 5 Session
Transport	↔	4 Transport
Internet	↔	3 Network
Network Access	↔	2 Data Link, 1 Physical

4. Types of Networks

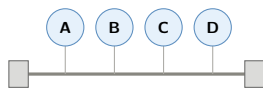
Networks are classified by their geographic scale:

Type	Range	Example
PAN	~10 m	Bluetooth headphones
LAN	100 m – 1 km	Home/office network
MAN	10 – 100 km	City cable infrastructure
WAN	Global	The Internet
WLAN	Variable	Wi-Fi network
VPN	Virtual	Encrypted tunnel over internet

5. Network Topologies

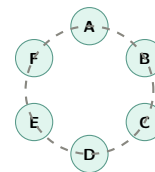
Bus

All devices share a single cable. Simple and cheap, but prone to collisions. Obsolete today.



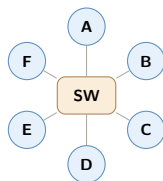
Ring

Each device connects to the next, forming a loop. Data travels in one direction via token passing. One failure can break the ring.



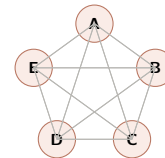
Star

All devices connect to a central hub or switch. Most common topology today. Easy to manage; hub is a single point of failure.



Mesh

Every device connects to every other. Highly redundant and fault-tolerant. Expensive to wire. Used in WANs and the internet backbone.



6. IP Addressing

6.1 IPv4

A 32-bit address written as four octets: 192.168.1.100



Class	Range	Mask	Use
A	1.0.0.0 -- 126.255.255.255	/8	Very large orgs
B	128.0.0.0 -- 191.255.255.255	/16	Medium orgs
C	192.0.0.0 -- 223.255.255.255	/24	Small orgs
D	224.0.0.0 -- 239.255.255.255	—	Multicast

10.x.x.x · 172.16.x.x -- 172.31.x.x · 192.168.x.x

6.2 IPv6

A 128-bit address: 2001:0db8:85a3::8a2e:0370:7334

- ~340 undecillion addresses (3.4×10^{38})
- Designed to replace the nearly-exhausted IPv4 space
- Supports auto-configuration (SLAAC), built-in IPsec

6.3 CIDR and Subnetting

CIDR (Classless Inter-Domain Routing) notation: 192.168.1.0/24

The **prefix length** (e.g. /24) indicates how many bits are the network portion.

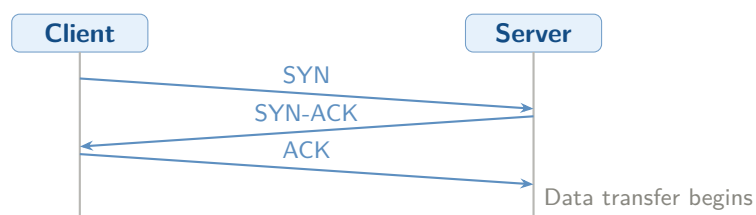
CIDR	Subnet Mask	Usable Hosts
/8	255.0.0.0	16,777,214
/16	255.255.0.0	65,534
/24	255.255.255.0	254
/25	255.255.255.128	126
/30	255.255.255.252	2

7. TCP vs UDP

Property	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed delivery	Best-effort
Ordering	In-order delivery	No guarantee
Error check	Yes + retransmit	Checksum only
Flow control	Yes	No
Speed	Slower (overhead)	Much faster
Header size	20–60 bytes	8 bytes
Use cases	Web, email, SSH, FTP	Video, gaming, DNS, VoIP

7.1 TCP 3-Way Handshake

Before data flows, TCP establishes a connection:



8. Key Protocols

8.1 Application Layer

Protocol	Port	Purpose
HTTP/S	80/443	Foundation of the web; HTTPS adds TLS encryption
DNS	53	Translates domain names to IP addresses
DHCP	67/68	Automatically assigns IP addresses to devices
FTP	20/21	File transfer between client and server
SMTP	25/587	Sending email; POP3/IMAP for receiving
SSH	22	Encrypted remote shell access; replaced Telnet
Telnet	23	Unencrypted remote access (obsolete/insecure)
SNMP	161	Network device monitoring and management
NTP	123	Time synchronisation across network devices

8.2 Network Layer

Protocol	Purpose
ARP	Resolves IP addresses to MAC addresses on local segment
ICMP	Diagnostics — powers ping and traceroute
NAT	Maps private IPs to a public IP at the router
OSPF	Link-state routing inside organisations (uses Dijkstra's)
BGP	Internet's routing protocol — exchanges routes between ISPs
RIP	Distance-vector routing (max 15 hops), used in small networks

9. Network Devices

Device	OSI Layer	Description
Hub	Layer 1	Broadcasts incoming data to ALL ports. Creates collision domains. Obsolete.
Switch	Layer 2	Learns MAC addresses; sends frames only to the correct port. Backbone of LANs.
Router	Layer 3	Forwards packets between different networks using IP and routing tables.
Firewall	L3–L7	Monitors and controls traffic by security rules; can do deep packet inspection.
Bridge	Layer 2	Connects two network segments; reduces collision domains.
Gateway	All	Translates between different protocols; your home router is your default gateway.
Modem	Layer 1–2	Modulates/demodulates signal to connect to the ISP over telephone/cable/fibre.

10. DNS — How Domain Names Work

When you type `google.com` in a browser, the following happens in milliseconds:

1. Browser checks its **local cache** — was this queried recently?
2. Asks the **Recursive Resolver** (e.g. your ISP or 8.8.8.8)
3. Resolver queries a **Root Name Server** — “who handles `.com`?”
4. Root refers to the **TLD Name Server** for `.com`
5. TLD refers to **Google's Authoritative Name Server**
6. Authoritative server returns the IP (e.g. 142.250.x.x)
7. Browser connects via TCP/TLS to that IP

DNS record types

A	Maps hostname to IPv4 address
AAAA	Maps hostname to IPv6 address
CNAME	Canonical name — alias from one name to another
MX	Mail exchange — specifies mail server for a domain
NS	Name server — delegates a subdomain to a nameserver
TXT	Text record — used for SPF, DKIM, domain verification

11. Network Security

11.1 Common Attacks

Attack	Description
DDoS	Overwhelms a server with traffic from thousands of sources
Man-in-the-Middle	Attacker intercepts communication between two parties
ARP Poisoning	Fake ARP replies redirect traffic to attacker's machine on LAN
DNS Spoofing	Returns false DNS records to redirect users to malicious sites
Port Scanning	Probing ports to discover open services and vulnerabilities
SYN Flood	Sends many SYN packets without completing handshakes, exhausting server

11.2 Defences

Defence	Description
TLS/SSL	Encrypts data in transit; HTTPS uses TLS over HTTP
Firewall	Filters traffic by IP, port, or application rules
VPN	Encrypts all traffic through a secure tunnel
IDS/IPS	Intrusion Detection/Prevention Systems monitor for suspicious patterns
Network segmentation	VLANs isolate critical systems so breaches don't spread
802.1X	Port-based authentication before granting network access

11.3 Encryption Basics

- **Symmetric encryption** — same key encrypts and decrypts (e.g. AES-256). Fast; used for bulk data.
- **Asymmetric encryption** — public key encrypts, private key decrypts (e.g. RSA-2048). Used in TLS handshake.
- **TLS handshake** — asymmetric crypto is used to exchange a symmetric session key; then symmetric takes over for speed.
- **Hashing** — one-way function (SHA-256) verifies integrity without revealing content.

11.4 Common Port Numbers

Port	Service	Port	Service	Port	Service
20/21	FTP	80	HTTP	443	HTTPS
22	SSH	110	POP3	3306	MySQL
23	Telnet	143	IMAP	3389	RDP
25	SMTP	161	SNMP	8080	HTTP-alt
53	DNS	179	BGP	27017	MongoDB
67/68	DHCP	389	LDAP	5432	PostgreSQL

12. Advanced Concepts

12.1 VLANs (Virtual LANs)

VLANs logically separate devices on the same physical switch. An accounting VLAN and an engineering VLAN can share the same hardware but remain completely isolated — critical for security and traffic management.

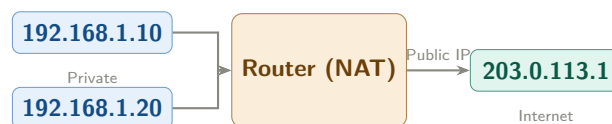
VLAN tagging (802.1Q) adds a 4-byte tag to Ethernet frames carrying the VLAN ID. Trunk ports carry multiple VLANs; access ports carry one.

12.2 Routing Algorithms

- **OSPF** (Open Shortest Path First) — Link-state protocol. Each router has the full network topology and computes shortest paths using Dijkstra's algorithm. Used *inside* organisations (IGP).
- **BGP** (Border Gateway Protocol) — Path-vector protocol. The internet's routing protocol. ISPs exchange reachability (AS paths). Controls how data traverses the global internet (EGP).
- **RIP** (Routing Information Protocol) — Distance-vector, older. Maximum 15 hops. Used in small or legacy networks.
- **EIGRP** — Cisco proprietary; hybrid distance-vector/link-state. Fast convergence.

12.3 NAT (Network Address Translation)

NAT allows many devices with private IPs to share a single public IP address. Your home router performs NAT — all your devices' traffic appears to the internet as a single IP.



12.4 QoS (Quality of Service)

QoS prioritises certain traffic types (VoIP, video calls) over others (file downloads) to ensure real-time applications perform well under congestion.

12.5 SDN (Software-Defined Networking)

SDN decouples the **control plane** (routing decisions) from the **data plane** (packet forwarding). A central controller programs network behaviour dynamically — enabling cloud-scale, programmable networks.

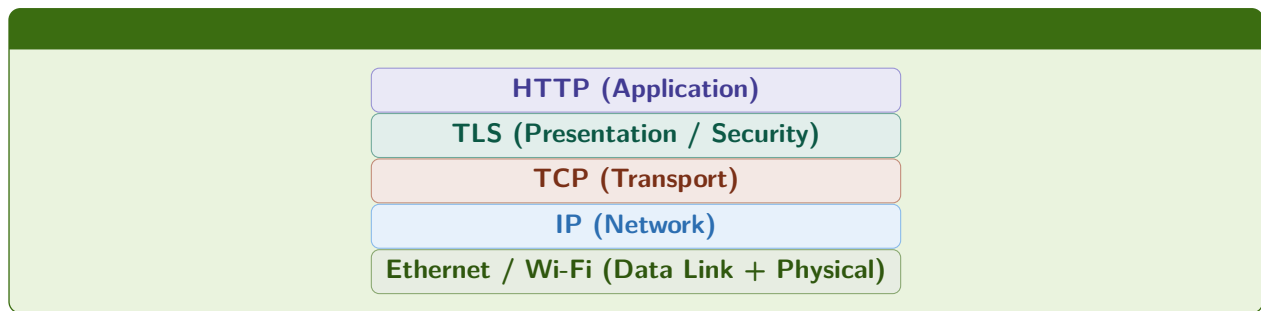
12.6 CDN (Content Delivery Network)

Geographically distributed servers cache content close to users. When you load a streaming video, you're typically hitting a CDN edge server nearby, not a central data centre thousands of kilometres away.

13. What Happens When You Visit a Website

A complete walkthrough of `https://example.com`:

1. You type `https://example.com` in your browser.
2. **DNS lookup** — `example.com` → IP address (UDP, port 53). Checks local cache → recursive resolver → root NS → TLD NS → authoritative NS.
3. **TCP handshake** (SYN → SYN-ACK → ACK) to the server on port 443.
4. **TLS handshake** — browser and server negotiate cipher suite, exchange certificates, derive session keys.
5. **HTTP GET** request sent over the encrypted TLS tunnel.
6. Server returns HTML (HTTP 200 OK) inside the encrypted TLS record.
7. Browser parses HTML; additional requests fire for CSS, JS, images — each may trigger new DNS lookups or reuse existing connections (HTTP/2 multiplexing).
8. Connection closes via TCP FIN handshake (or stays alive for keep-alive).



13. Quick Cheat Sheet

OSI layers (7 → 1):

Application · Presentation · Session
Transport · Network · Data Link · Physical

TCP/IP (4 → 1):

Application · Transport · Internet · Network Access

TCP = reliable, ordered, connection-oriented

UDP = fast, connectionless, best-effort

IPv4: 32-bit, dotted decimal, $\sim 4.3\text{B}$ addresses

IPv6: 128-bit, hex colon, $\sim 3.4 \times 10^{38}$ addresses

Key devices:

Hub (L1) · Switch (L2) · Router (L3) · Firewall

Routing protocols:

OSPF (interior, link-state)
BGP (inter-AS, path-vector)
RIP (interior, distance-vector)

Must-know ports:

22 SSH · 53 DNS · 80 HTTP · 443 HTTPS
25 SMTP · 67 DHCP · 3389 RDP

ARP: IP → MAC **DNS:** name → IP

NAT: private IP → public IP